Polynomial End Behavior & Zeros

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Leading Term Test and End Behavior**

A polynomial $f(x)$ of degree 4 with real coefficients has leading coefficient 2 and zeros at $x = -1$ (multiplicity 2) and $x = 2 + i$. Write $f(x)$ in fully expanded standard form.

Step 1: Apply the Conjugate Root Theorem: Since $f(x)$ has real coefficients, the complex conjugate $2 - i$ must also be a zero. Step 2: Construct the factored form: $f(x) = 2(x+1)^2((x-2)-i)((x-2)+i) = 2(x^2+2x+1)((x-2)^2+1)$. Step 3: Expand and simplify: $(x-2)^2+1 = x^2-4x+5$.

$f(x) = 2(x^2+2x+1)(x^2-4x+5) = 2(x^4-2x^3-2x^2+6x+5) = \mathbf{2x^4-4x^3-4x^2+12x+10}$.

2 **[Foundational]** • **Target: Multiplicity of Zeros and Graph Geometry**

When the polynomial $P(x) = 2x^3 + kx^2 - 7x + 6$ is divided by $(x-2)$, the remainder is 12. Find the value of k, and determine whether $(x+3)$ is a factor of $P(x)$.

Step 1: Apply the Remainder Theorem $P(2) = 12$: $2(2)^3 + k(2)^2 - 7(2) + 6 = 12 \Rightarrow 16 + 4k - 14 + 6 = 12 \Rightarrow 4k + 8 = 12 \Rightarrow \mathbf{k = 1}$. Step 2: Test $(x+3)$ using the Factor Theorem: With $k = 1$, $P(x) = 2x^3 + x^2 - 7x + 6$. Evaluate $P(-3)$: $P(-3) = 2(-27) + (-3)^2 - 7(-3) + 6 = -54 + 9 + 21 + 6 = -18 eq 0$. Since $P(-3) = -18 eq 0$, $(x+3)$ is not a factor.

3 **[Foundational]** • **Target: Remainder & Factor Theorems**

Determine the end behavior of $P(x) = -4x^5 + 7x^3 - 2x + 9$ as $x \to -\infty$ and as $x \to +\infty$ using limit notation, and state the maximum number of local extrema.

Step 1: Identify leading term: The leading term is $-4x^5$, which has odd degree $n = 5$ and negative leading coefficient $a_5 = -4 < 0$. Step 2: Write limit statements: $\lim_{x \to -\infty} P(x) = +\infty$, $\lim_{x \to +\infty} P(x) = -\infty$. Step 3: Bound on local extrema: A polynomial of degree $n = 5$ has at most $n - 1 = \mathbf{4}$ local extrema (turning points).

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4 [Foundational] • Target: Rational Root Theorem & Complex Conjugates

The graph of the exponential function $f(x) = a \cdot \log_3(x - h) + k$ is shown below. The function has a vertical asymptote at $x = 3$ and passes through the points $(4, 5)$ and $(12, 7)$. Determine the value of the parameter a.

The general form of the logarithmic function is $f(x) = a \cdot \log_b(x - h) + k$. The vertical asymptote occurs at $x = h$, so we are given that $h = 3$. Thus, the function can be written as $f(x) = a \cdot \log_3(x - 3) + k$. We are given two points on the graph: $(4, 5)$ and $(12, 7)$. Substituting these points into the function, we get a system of two equations: 1) $5 = a \cdot \log_3(4 - 3) + k$ $5 = a \cdot \log_3(1) + k$ Since $\log_3(1) = 0$, this simplifies to $5 = k$. 2) $7 = a \cdot \log_3(12 - 3) + k$ $7 = a \cdot \log_3(9) + k$ Since $\log_3(9) = 2$, this becomes $7 = 2a + k$. Now substitute the value of k from the first equation into the second equation: $7 = 2a + 5$ Subtract 5 from both sides: $2 = 2a$ Divide by 2: $a = 1$. Therefore, the value of the parameter a is 1.

5 [Foundational] • Target: Leading Term Test and End Behavior

The graph of the quadratic function $f(x) = x^2 - bx + c$ is shown below. The vertex of the parabola is at $(2, -3)$, and it passes through the point $(0, 1)$. Determine the value of the parameter b.

The standard form of a quadratic function is $f(x) = ax^2 + bx + c$. In this problem, $a = 1$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -3)$ and $a = 1$, the function can be written as $f(x) = 1(x - 2)^2 - 3$. Expanding this, we get $f(x) = (x^2 - 4x + 4) - 3 = x^2 - 4x + 1$. We are given that $f(x) = x^2 - bx + c$. By comparing the coefficients, we see that $-b = -4$, which implies $b = 4$, and $c = 1$. We can verify this by checking if the point $(0, 1)$ lies on the graph of $f(x) = x^2 - 4x + 1$. Substituting $x = 0$, we get $f(0) = 0^2 - 4(0) + 1 = 1$, which matches the given point. Therefore, the value of b is 4.

6 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

For the rational function $r(x) = \frac{2x^2 - 2x - 12}{x^2 - 9}$, find: (a) the domain, (b) the coordinates of any removable discontinuity (hole), (c) the vertical and horizontal asymptotes, and (d) the x- and y-intercepts.

Step 1: Factor numerator and denominator completely: $r(x) = \frac{2(x^2 - x - 6)}{(x - 3)(x + 3)} = \frac{2(x - 3)(x + 2)}{(x - 3)(x + 3)}$. Step 2: Answer each part:(a) Domain: $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$, i.e. $x \neq \pm 3$.(b) Removable Discontinuity: The factor $(x - 3)$ cancels; for $x \neq 3$, $r(x) = \frac{2(x + 2)}{x + 3}$. Evaluating at $x = 3$ gives $y = \frac{2(5)}{6} = \frac{5}{3}$. Hole at $\left(\mathbf{3, \frac{5}{3}}\right)$.(c) Asymptotes: Vertical asymptote at $\mathbf{x = -3}$; horizontal asymptote at $\mathbf{y = 2}$.(d) Intercepts: x-intercept at $\mathbf{(-2, 0)}$; y-intercept at $\left(\mathbf{0, \frac{4}{3}}\right)$.

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7 [Foundational] • Target: Remainder & Factor Theorems

Consider the quadratic function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

To find the x-intercepts of the graph of a function, we need to find the values of x for which $f(x) = 0$. For the given function $f(x) = x^2 - 4x + 3$, we set $x^2 - 4x + 3 = 0$. This is a quadratic equation that can be solved by factoring. We look for two numbers that multiply to 3 and add to -4. These numbers are -1 and -3. Thus, we can factor the equation as $(x - 1)(x - 3) = 0$. Setting each factor to zero, we get $x - 1 = 0$ or $x - 3 = 0$. Solving these equations gives $x = 1$ and $x = 3$. Therefore, the x-intercepts are $(1, 0)$ and $(3, 0)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

Consider the polynomial function $f(x) = x^2 - 5x + 6$. What are the roots of the function $f(x)$?

To find the roots of the function $f(x) = x^2 - 5x + 6$, we need to solve the equation $f(x) = 0$. This means we need to find the values of x for which $x^2 - 5x + 6 = 0$. This is a quadratic equation that can be solved by factoring. We are looking for two numbers that multiply to 6 and add up to -5. These numbers are -2 and -3. Therefore, we can factor the quadratic as $(x - 2)(x - 3) = 0$. Setting each factor to zero gives us the roots: $x - 2 = 0 \Rightarrow x = 2$ and $x - 3 = 0 \Rightarrow x = 3$. Thus, the roots of the function $f(x)$ are 2 and 3.

9 [Intermediate] • Target: Leading Term Test and End Behavior

The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P?

The function is given by $f(x) = 2^x$. We are looking for a point P on the graph where the y-coordinate is 8. This means we need to solve for x when $f(x) = 8$. So, we have the equation $2^x = 8$. Since $8 = 2^3$, we can rewrite the equation as $2^x = 2^3$. By equating the exponents, we find that $x = 3$. Therefore, the x-coordinate of point P is 3.

10 [Intermediate] • Target: Multiplicity of Zeros and Graph Geometry

Find the average rate of change of the polynomial function $f(x) = 2x^3 - 5x + 4$ over the closed interval $[-1, 2]$.

Step 1: Evaluate the function at the interval endpoints: $f(-1) = 2(-1)^3 - 5(-1) + 4 = -2 + 5 + 4 = 7$. $f(2) = 2(2)^3 - 5(2) + 4 = 16 - 10 + 4 = 10$. Step 2: Compute the difference quotient: $\text{AROC} = \frac{f(2) - f(-1)}{2 - (-1)} = \frac{10 - 7}{3} = \frac{3}{3} = \mathbf{1}$.

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11 **[Intermediate]** • **Target: Remainder & Factor Theorems**

Solve the rational inequality $\frac{x+4}{x-2} \leq 3$ and express the solution set in interval notation.

Step 1: Subtract 3 and combine over a single common denominator: $\frac{x+4}{x-2} - 3 \leq 0 \Rightarrow \frac{(x+4)-3(x-2)}{x-2} \leq 0 \Rightarrow \frac{-2x+10}{x-2} \leq 0$. Step 2: Factor and identify critical boundary values: $\frac{-2(x-5)}{x-2} \leq 0 \Leftrightarrow \frac{x-5}{x-2} \geq 0$. Critical values are $x = 2$ (undefined) and $x = 5$ (zero). Step 3: Sign chart analysis: The quotient $\frac{x-5}{x-2}$ is positive on $(-\infty, 2)$ and $(5, \infty)$, and negative on $(2, 5)$. Including the zero at $x = 5$ and excluding $x = 2$: $\mathbf{(-\infty, 2) \cup [5, \infty)}$.

12 **[Intermediate]** • **Target: Rational Root Theorem & Complex Conjugates**

Find the equation of the slant (oblique) asymptote of the rational function $g(x) = \frac{3x^2 - 5x + 7}{x - 2}$, and compute $\lim_{x \to 2^+} g(x)$.

Step 1: Perform polynomial long division (or synthetic division by $c = 2$): $3x^2 - 5x + 7 = (x - 2)(3x + 1) + 9 \Rightarrow g(x) = 3x + 1 + \frac{9}{x-2}$. Step 2: Identify the slant asymptote: As $x \to \pm\infty$, $\frac{9}{x-2} \to 0$, so the slant asymptote is $\mathbf{y = 3x + 1}$. Step 3: One-sided limit at $x = 2^+$: As $x \to 2^+$, $x - 2 \to 0^+$ and numerator $\to 9 > 0$, so $\lim_{x \to 2^+} g(x) = \mathbf{+\infty}$.

13 **[Intermediate]** • **Target: Leading Term Test and End Behavior**

The graph of $y = f(x)$ is transformed into $y = -2f(3x - 6) + 5$. Describe the exact sequence of geometric transformations applied to $f(x)$, and find the image of the point $(3, -1)$ on $y = f(x)$.

Step 1: Factor the horizontal coefficient: $y = -2f(3(x - 2)) + 5$. Step 2: Sequence of transformations:Horizontal dilation (compression) by factor $\frac{1}{3}$.Horizontal translation 2 units right.Vertical dilation (stretch) by factor 2 and reflection across the x-axis.Vertical translation 5 units up. Step 3: Map the point $(3, -1)$: Set $3x - 6 = 3 \Rightarrow x = 3$. Then $y = -2f(3) + 5 = -2(-1) + 5 = 7$. Image point: **(3, 7)**.

14 **[Intermediate]** • **Target: Multiplicity of Zeros and Graph Geometry**

Let $z_1 = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ and $z_2 = 3\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$. Find the product $z_1 z_2$ in rectangular form $a + bi$, and compute z_1^6 using De Moivre's Theorem.

Step 1: Multiply in polar form: $z_1 z_2 = (2 \cdot 3)\left[\cos\left(\frac{\pi}{4} + \frac{5\pi}{12}\right) + i\sin\left(\frac{\pi}{4} + \frac{5\pi}{12}\right)\right] = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = \mathbf{-3 + 3i\sqrt{3}}$. Step 2: Compute z_1^6 via De Moivre's Theorem: $z_1^6 = 2^6\left(\cos\frac{6\pi}{4} + i\sin\frac{6\pi}{4}\right) = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = \mathbf{-64i}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Solve the polynomial inequality $x^3 - 4x^2 + x + 6 \leq 0$.

Step 1: Find rational roots and factor: Testing $x = -1$: $(-1)^3 - 4(-1)^2 + (-1) + 6 = -1 - 4 - 1 + 6 = 0$. Dividing by $(x + 1)$ gives: $x^3 - 4x^2 + x + 6 = (x + 1)(x^2 - 5x + 6) = (x + 1)(x - 2)(x - 3)$. Step 2: Sign analysis across roots $-1, 2, 3$:$x < -1$: $(-)(-)(-) < 0$$-1 < x < 2$: $(+)(-)(-) > 0$$2 < x < 3$: $(+)(+)(-) < 0$$x > 3$: $(+)(+)(+) > 0$Including zeros gives the solution set: $\mathbf{(-\infty, -1] \cup [2, 3]}$.

16 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 4.01 Inverse & Composite Functions

A geometric sequence has third term $g_3 = 18$ and sixth term $g_6 = -\frac{16}{3}$. Find the common ratio r, the first term g_1, and the infinite sum S_∞.

Step 1: Find common ratio r: $\frac{g_6}{g_3} = r^3 = \frac{-16/3}{18} = -\frac{8}{27} \Rightarrow r = \mathbf{-\frac{2}{3}}$. Step 2: Find first term g_1: $g_1 r^2 = 18 \Rightarrow g_1\left(\frac{4}{9}\right) = 18 \Rightarrow g_1 = \mathbf{\frac{81}{2}}$. Step 3: Infinite geometric series sum ($|r| = 2/3 < 1$): $S_\infty = \frac{g_1}{1 - r} = \frac{81/2}{1 - (-2/3)} = \frac{81/2}{5/3} = \mathbf{\frac{243}{10} = 24.3}$.

17 [Challenging] • Target: Leading Term Test and End Behavior • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A log-log regression plot of $(\ln x, \ln y)$ produces the linear fit $\ln(\hat{y}) = \ln(5) - 2.5\ln(x)$ for $x > 0$. Write $\hat{y}$ as an explicit power function of x, and find the factor by which $\hat{y}$ changes when x is multiplied by 4.

Step 1: Condense the right-hand side using logarithm power and quotient laws: $\ln(\hat{y}) = \ln(5) + \ln(x^{-2.5}) = \ln(5x^{-5/2}) \Rightarrow \mathbf{\hat{y} = 5x^{-5/2} = \frac{5}{\sqrt{x^5}}}$. Step 2: Scaling analysis when $x \to 4x$: $\hat{y}(4x) = 5(4x)^{-5/2} = 4^{-5/2} \cdot 5x^{-5/2} = \frac{1}{(\sqrt{4})^5}\hat{y}(x) = \mathbf{\frac{1}{32}}\hat{y}(x)$.

18 [Challenging] • Target: Multiplicity of Zeros and Graph Geometry • Cross-Lesson: 4.03 Parametric Equations & Modeling

Use the Binomial Theorem to find the coefficient of x^3y^2 in the expansion of $(2x - 3y)^5$.

Step 1: Apply the general binomial term formula $\binom{n}{k}a^{n-k}b^k$: With $n = 5$, $a = 2x$, $b = -3y$, and $k = 2$: $T_3 = \binom{5}{2}(2x)^3(-3y)^2$. Step 2: Evaluate numerical factors: $\binom{5}{2} = 10, \quad (2x)^3 = 8x^3, \quad (-3y)^2 = 9y^2$. $T_3 = 10 \cdot 8 \cdot 9\, x^3y^2 = 720x^3y^2$. The coefficient is $\mathbf{720}$.

19 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the 12th term and the sum of the first 25 terms of the arithmetic sequence with $a_1 = -7$ and common difference $d = 4$.

Step 1: General term formula $a_n = a_1 + (n - 1)d$: $a_{12} = -7 + (12 - 1)(4) = -7 + 44 = \mathbf{37}$. Step 2: Arithmetic series sum $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$: $S_{25} = \frac{25}{2}(2(-7) + 24(4)) = \frac{25}{2}(-14 + 96) = \frac{25}{2}(82) = 25 \times 41 = \mathbf{1025}$.

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20 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 1.03 Rates of Change & Concavity

A point P is on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. The coordinates of P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. If a point Q on the unit circle has an angle ϕ such that $\phi = \theta + \frac{\pi}{3}$, what is the y-coordinate of point Q?

The coordinates of point P on the unit circle are given as $(\cos\theta, \sin\theta)$. Therefore, $\cos\theta = \frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$. This implies that $\theta = \frac{\pi}{3}$ (or 60 degrees). The point Q on the unit circle has an angle $\phi = \theta + \frac{\pi}{3}$. Substituting the value of θ, we get $\phi = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$ (or 120 degrees). The coordinates of Q are $(\cos\phi, \sin\phi)$. We need to find the y-coordinate of Q, which is $\sin\phi$. Therefore, the y-coordinate of Q is $\sin\left(\frac{2\pi}{3}\right)$. Using the unit circle or trigonometric identities, $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Leading Term Test and End Behavior

Determine the end behavior of the polynomial function $f(x) = -3x^5 + 7x^3 - 2x + 9$.

a. **As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$** ✓

b. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$

c. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$

d. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$

The leading term is $-3x^5$. Because the degree 5 is odd and the leading coefficient -3 is negative, $f(x) \to -\infty$ as $x \to \infty$, and $f(x) \to \infty$ as $x \to -\infty$.

22 [Foundational] • Target: Leading Term Test and End Behavior

If a polynomial $p(x)$ satisfies $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = -\infty$, which of the following could be the leading term of $p(x)$?

a. $-4x^6$ ✓

b. $2x^6$

c. $-5x^7$

d. $3x^5$

When both ends approach $-\infty$, the polynomial must have an even degree and a negative leading coefficient. Among the choices, $-4x^6$ has even degree 6 and negative coefficient -4.

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23 [Foundational] • Target: Leading Term Test and End Behavior

Consider $g(x) = (2 - x)(x + 3)^2(2x - 1)$. What is the leading coefficient of $g(x)$?

a. -2 ✓

b. 2

c. -1

d. 4

Expanding the highest power terms: $(-x) \cdot (x)^2 \cdot (2x) = -2x^4$. Thus, the leading coefficient is -2.

24 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

Let $f(x) = (x - 4)^3(x + 2)^2(x - 1)$. At which zero does the graph of f touch the x-axis and turn around without crossing?

a. $x = -2$ ✓

b. $x = 4$

c. $x = 1$

d. $x = 2$

A graph touches the x-axis and turns around at zeros with even multiplicity. Here, $(x + 2)^2$ has multiplicity 2 (even), so $x = -2$ is a turning touch point. At $x = 4$ and $x = 1$, the graph crosses.

25 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

A polynomial $p(x)$ of degree 4 has a zero at $x = 3$ of multiplicity 2, and no other real zeros. If $p(0) = 18$, which equation represents $p(x)$ if its remaining zeros are non-real complex conjugates $\pm i$?

a. $p(x) = 2(x - 3)^2(x^2 + 1)$ ✓

b. $p(x) = (x - 3)^2(x^2 + 1)$

c. $p(x) = 2(x - 3)^2(x^2 - 1)$

d. $p(x) = \frac{1}{2}(x - 3)^2(x^2 + 1)$

Zeros at $\pm i$ correspond to factor $(x^2 + 1)$. So $p(x) = a(x - 3)^2(x^2 + 1)$. Using $p(0) = a(-3)^2(1) = 9a = 18 \Rightarrow a = 2$. Thus $p(x) = 2(x - 3)^2(x^2 + 1)$.

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26 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

At $x = c$, $f(x) = (x - c)^3 g(x)$ where $g(c) \neq 0$. What is the behavior of the graph of f at $(c, 0)$?

a. The graph crosses the x-axis with an inflection/flattening point (horizontal tangent) ✓

b. The graph touches the x-axis and turns around symmetrically

c. The graph has a vertical asymptote at $x = c$

d. The graph crosses the x-axis linearly without flattening

An odd multiplicity greater than 1 ($m = 3$) causes the graph to cross the x-axis while flattening out (creating an inflection point with a horizontal tangent).

27 [Foundational] • Target: Remainder & Factor Theorems

When $f(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x - 3)$, the remainder is 7. Find the value of k.

a. -4 ✓

b. 4

c. -2

d. 5

By the Remainder Theorem, $f(3) = 7$. $f(3) = 2(27) - 5(9) + 3k - 8 = 54 - 45 + 3k - 8 = 1 + 3k$. Setting $1 + 3k = 7 \Rightarrow 3k = 6 \Rightarrow k = 2$.

28 [Foundational] • Target: Remainder & Factor Theorems

Which of the following is a factor of $P(x) = x^3 - 6x^2 + 11x - 6$?

a. $x - 2$ ✓

b. $x + 2$

c. $x - 4$

d. $x + 1$

Evaluate $P(2) = 2^3 - 6(4) + 11(2) - 6 = 8 - 24 + 22 - 6 = 0$. Since $P(2) = 0$, by the Factor Theorem $(x - 2)$ is a factor.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Remainder & Factor Theorems

A polynomial $q(x)$ satisfies $q(-4) = 0$, $q(1) = 0$, and $q(5) = 12$. Which statement must be true?

a. $(x + 4)$ **and** $(x - 1)$ **are factors of** $q(x)$ ✓

b. $(x - 4)$ and $(x + 1)$ are factors of $q(x)$

c. The remainder when $q(x)$ is divided by $(x + 5)$ is 12

d. $q(x)$ must be a quadratic polynomial

By the Factor Theorem, $q(c) = 0 \Leftrightarrow (x - c)$ is a factor. Here $q(-4) = 0 \Rightarrow (x + 4)$ is a factor, and $q(1) = 0 \Rightarrow (x - 1)$ is a factor.

30 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

According to the Rational Root Theorem, which of the following is NOT a possible rational zero of $f(x) = 3x^3 - 2x^2 + 5x - 4$?

a. $\frac{3}{2}$ ✓

b. $\frac{4}{3}$

c. ± 1

d. ± 2

Factors of constant term -4: $\pm 1, \pm 2, \pm 4$. Factors of leading coefficient 3: $\pm 1, \pm 3$. Possible rational roots p/q are $\pm 1, \pm 2, \pm 4, \pm 1/3, \pm 2/3, \pm 4/3$. Thus $\frac{3}{2}$ is not possible.

31 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

A polynomial with real coefficients has degree 3. Two of its zeros are 4 and $2 - 3i$. What is the third zero?

a. $2 + 3i$ ✓

b. $-2 + 3i$

c. -4

d. $-2 - 3i$

By the Complex Conjugate Root Theorem, non-real complex zeros of polynomials with real coefficients must occur in conjugate pairs. Since $2 - 3i$ is a zero, $2 + 3i$ must also be a zero.

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32 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

What is the total number of complex zeros (counting multiplicity) of $P(x) = (x^2 + 4)^2(x - 5)^3$?

a. 7 ✓

b. 5

c. 4

d. 3

The degree of $P(x)$ is $2 \times 2 + 3 = 7$. By the Fundamental Theorem of Algebra, a polynomial of degree n has exactly n complex zeros counting multiplicity.

33 [Intermediate] • Target: Leading Term Test and End Behavior

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. The graph of $P(x)$ has a y-intercept at $(0, 12)$. It is known that $P(x)$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. If the value of the local maximum is 20, determine the value of the constant term d of the polynomial.

a. **12** ✓

b. 10

c. 8

d. 6

Let the cubic polynomial be $P(x) = ax^3 + bx^2 + cx + d$. \\ We are given that the y-intercept is at $(0, 12)$, so $P(0) = 12$. \\ Substituting $x = 0$ into the polynomial, we get $P(0) = a(0)^3 + b(0)^2 + c(0) + d = d$. \\ Therefore, $d = 12$. \\ The problem statement asks for the value of the constant term d, which is directly given by the y-intercept.

34 [Intermediate] • Target: Multiplicity of Zeros and Graph Geometry

For the polynomial $f(x) = -1x^3 + 3x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. **up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)** ✓

d. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

The leading term is $-1x^3$, degree odd, leading coefficient negative, so the end behavior is: up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$).

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35 [Intermediate] • Target: Remainder & Factor Theorems

For the polynomial $f(x) = -2x^3 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓

b. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

The leading term is $-2x^3$, degree odd, leading coefficient negative, so the end behavior is: up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$).

36 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

For the polynomial $f(x) = -3x^5 + 3x - 1$, describe the end behavior of f.

a. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

b. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓

c. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

The leading term is $-3x^5$, degree odd, leading coefficient negative, so the end behavior is: up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$).

37 [Intermediate] • Target: Leading Term Test and End Behavior

For the polynomial $f(x) = -1x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓

b. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

c. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

The leading term is $-1x^5$, degree odd, leading coefficient negative, so the end behavior is: up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$).

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiplicity of Zeros and Graph Geometry • Cross-Lesson: 4.03 Parametric Equations & Modeling

For the polynomial $f(x) = -2x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓

b. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

The leading term is $-2x^5$, degree odd, leading coefficient negative, so the end behavior is: up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$).

39 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 1.02 Rational Functions & Asymptotes

For the polynomial $f(x) = 2x^3 + 3x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$) ✓

b. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

c. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

d. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

The leading term is $2x^3$, degree odd, leading coefficient positive, so the end behavior is: down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$).

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40 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 1.03 Rates of Change & Concavity

For the polynomial $f(x) = 3x^5 + 3x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$) ✓

b. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

The leading term is $3x^5$, degree odd, leading coefficient positive, so the end behavior is: down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$).

41 [Challenging] • Target: Leading Term Test and End Behavior • Cross-Lesson: 2.01 Exponential Functions & Growth Models

For the polynomial $f(x) = 2x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

b. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$) ✓

The leading term is $2x^5$, degree odd, leading coefficient positive, so the end behavior is: down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$).

42 [Challenging] • Target: Multiplicity of Zeros and Graph Geometry • Cross-Lesson: 2.02 Logarithmic Functions & Properties

For the polynomial $f(x) = 1x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

b. down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$) ✓

The leading term is $1x^5$, degree odd, leading coefficient positive, so the end behavior is: down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$).

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43 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

For the polynomial $f(x) = -1x^2 + 2x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

b. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

c. **down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓**

d. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

The leading term is $-1x^2$, degree even, leading coefficient negative, so the end behavior is: down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$).

44 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

For the polynomial $f(x) = -3x^2 + 2x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

b. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

c. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. **down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓**

The leading term is $-3x^2$, degree even, leading coefficient negative, so the end behavior is: down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$).

45 [Challenging] • Target: Leading Term Test and End Behavior • Cross-Lesson: 4.01 Inverse & Composite Functions

For the polynomial $f(x) = -2x^2 + 2x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to +\infty$)

b. **down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$) ✓**

c. up on both ends (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to +\infty$)

d. up on the left, down on the right (as $x \to -\infty, f(x) \to +\infty$; as $x \to +\infty, f(x) \to -\infty$)

The leading term is $-2x^2$, degree even, leading coefficient negative, so the end behavior is: down on both ends (as $x \to -\infty, f(x) \to -\infty$; as $x \to +\infty, f(x) \to -\infty$).

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